



EAD Variance Analyzer with LangChain

Upload two datasets (old vs new) to analyze EAD changes and get an AI explanation.

Upload OLD dataset (xlsx)



Drag and drop file here
Limit 200MB per file • XLSX

Browse files



inputs_data1_v4.xlsx 22.9KB



Upload NEW dataset (xlsx)



Drag and drop file here
Limit 200MB per file • XLSX

Browse files



inputs_data2_v4.xlsx 24.1KB



Conversation Controls



Start New Conversation

Ask a Question



if netting set MNA004 is unmargined, what will be the EAD change?



Impact Analysis: MNA004 Changing from Margined to Unmargined

Current State (Margined)

Based on the EAD comparison data, netting set **MNA004** currently has:

- **Margin Status:** Margined
- **Replacement Cost (RC):** 30,289.0
- **PFE Add-on:** 424,977.81
- **Multiplier:** 1.0
- **Current EAD:** 637,373.54

Expected Changes if Unmargined

If MNA004 were to become **unmargined**, the following components would be affected:

1. Maturity Factor (MF) Change

- **Current (Margined):** $MF = 1.5 \times \sqrt{(MPOR/250)}$
- **If Unmargined:** $MF = \sqrt{(\min(\max(\text{Maturity}, 10), 250)/250)}$

For unmargined netting sets, the maturity factor calculation shifts from being MPOR-based to maturity-based. This typically results in a **higher maturity factor** for trades with longer maturities, as the formula uses the actual trade maturity (capped between 10 and 250 days) rather than the margin period of risk.

2. PFE Add-on Impact

Since the maturity factor feeds into the trade add-on calculation:

- **Trade Add-on** = Adjusted Notional × **Maturity Factor** × Supervisory Delta × Supervisory Factor

A higher maturity factor would increase individual trade add-ons, which aggregate up through hedging sets to increase the **total PFE add-on** at the netting set level.

3. Replacement Cost (RC) Calculation

The RC formula also changes:

- **Current (Margined)**: $RC = \max(V - C, VMT + MTA - NICA, 0)$
- **If Unmargined**: $RC = \max(V - C, 0)$

For MNA004, the current RC is 30,289. If unmargined, the RC would be calculated as $\max(V - NICA, 0)$, where V is the sum of market values. Without collateral adjustments from VM, VMT, and MTA, the RC could potentially **increase** depending on the net market value position.

4. Multiplier Effect

The multiplier formula remains the same: $\min(1, 0.05 + 0.95 \times \exp((V-C)/(1.9 \times \text{AddOn})))$

However, with:

- Potentially higher RC (no VM offset)
- Higher PFE add-on (due to maturity factor increase)

The multiplier is currently at 1.0, so it would likely remain at or near 1.0.

Estimated EAD Impact

Expected Direction: EAD would INCREASE significantly

Magnitude Estimate:

- If maturity factors increase by 50-100% (typical for trades with maturities >20 days when moving from margined to unmargined), the PFE add-on could increase from ~425K to potentially **600K-850K**
- RC could increase if the netting set has positive net market value that was previously offset by VM
- Combined effect: **EAD could increase by 30-80%**, potentially reaching **800K-1.15M** (vs. current 637K)

Key Drivers of Change

1. **Maturity Factor recalculation** (MPOR-based → Maturity-based) - **Primary driver**
2. **Loss of VM collateral offset** in RC calculation - **Secondary driver**
3. **Longer effective holding period** reflected in risk metrics

Recommendation

If MNA004 is being considered for unmargined status, the bank should expect a material increase in EAD and corresponding capital requirements. The exact impact would depend on the specific maturity profile of trades within the netting set.

Enter a question...

